USN | | | | | | | | | | | [23BSMA23]

SECOND SEMESTER B.Tech EXAMINATION, JUNE - 2024

PARTIAL DIFFERENTIAL EQUATIONS AND LAPLACE TRANSFORMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**05x02=10**
1.	**a.**	Show that $\int_0^1\int_0^2 y^2 x\,dy\,dx = \frac{4}{3}$.	L2	CO1	**02**
	b.	Find the Laplace transform of e^{8t}.	L3	CO3	**02**
	c.	Find the Inverse Laplace transform of $\frac{1}{s+2}+\frac{1}{s-3}$.	L3	CO4	**02**
	d.	Solve by direct integration $\frac{\partial u}{\partial x} = x + y$.	L3	CO5	**02**
	e.	Write befitting solution for one dimensional heat and wave equations.	L3	CO6	**02**
		PART-B			**06x05=30**
2.	**a.**	Evaluate $\int_0^1\int_x^{\sqrt{x}} xy\,dy\,dx$.	L2	CO1	**05**
		OR			
	b.	Evaluate $\int_{-2}^{2}\int_0^{\sqrt{4-x^2}} (2-x)\,dy\,dx$ by changing the order of integration.	L2	CO1	**05**
3.	**a.**	Find the Laplace transform of $f(t) = \left[(3t+4)^3 + 5^t\right]$	L3	CO3	**05**
		OR			
	b.	Find the Laplace transform of $f(t) = \frac{\cos at - \cos bt}{t}$.	L3	CO3	**05**
4.	**a.**	Find the inverse Laplace transform of $\frac{(s+2)^3}{s^6}$.	L3	CO4	**05**
		OR			
	b.	Find the inverse Laplace transform of $\frac{2s-1}{s^2+4s+29}$.	L3	CO4	**05**
5.	**a.**	Solve by direct integration $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x+3y)$.	L3	CO5	**05**

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		OR			
	b.	Solve $p\cot x + q\cot y = \cot z$.	L3	CO5	**05**
6.	**a.**	Solve by the method of separation of variables $6\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 4u$.	L3	CO6	**05**
		OR			
	b.	Find one possible solution of one dimensional heat equation.	L3	CO6	**05**
7.	**a.**	Write MATLAB code for evaluation of double integrals with changing the order of integration.	L2	CO1	**05**
		OR			
	b.	Write any five MATLAB commands with explanation for Laplace Transforms.	L3	CO3	**05**
		PART - C		**06x10=60**	
8.	**a.**	Evaluate $\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z}\, dz\, dy\, dx$	L3	CO2	**10**
		OR			
	b.	A pyramid is bounded by three co-ordinate planes and the plane $x + 2y + 3z = 6$. Compute the volume by double integration.	L3	CO2	**10**
9.	**a.**	Define periodic function and find the Laplace transform of the periodic function defined by $f(t) = t^2, 0 < t < 2; f(t+2) = f(t)$.	L3	CO3	**10**
		OR			
	b.	If $f(t) = \begin{cases} \cos t, & 0 < t < \pi \\ \cos 2t, & \pi < t < 2\pi \\ \cos 3t, & t > 2\pi \end{cases}$ Express $f(t)$ in terms of unit step function and hence find its Laplace transform.	L3	CO3	**10**
10.	**a.**	Verify convolution theorem for the functions $f(t) = t$ *and* $g(t) = \cos t$	L3	CO4	**10**
		OR			
	b.	Use the Laplace transform to solve the differential equation $y''' + 2y'' - y' - 2y = 0$ *given* $y(0) = y'(0) = 0$ *and* $y''(0) = 6$	L3	CO4	**10**

11.	**a.**	*Solve* $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$ *for which* $\frac{\partial z}{\partial y} = -2\sin y$ *when* $x = 0$ *and* $z = 0$ *if* y *is an odd multiple of* $\pi/2$	L3	CO5	**10**
		OR			
	b.	Solve by the method of separation of variables $4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$, given that $u(0,y) = 2e^{5y}$	L3	CO5	**10**
12.	**a.**	Find the various possible solutions of one dimensional wave equation by variable separable method.	L3	CO6	**10**
		OR			
	b.	Obtain the solution of the heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ *subject to the conditions* $u(0,t) = 0$, $u(l,t) = 0$, *and* $u(x,0) = f(x)$	L3	CO6	**10**
13.	**a.**	Write MATLAB code for finding solution of ordinary differential equations using Laplace Transforms.	L3	CO4	**10**
		OR			
	b.	Obtain the D'Alembert's solution of one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ *subject to the conditions* $u(x,0) = f(x)$ *and* $\frac{\partial u}{\partial t}(x,0) = 0.$	L3	CO6	**10**